

Multiple Testing in Clinical Trials

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Protecting the overall type I error rate

Example

In a randomised clinical trial, two doses of a new treatment are compared against an active control.

Two null hypotheses are defined

$H_{0,1}$: No difference between dose 1 and the control,

$H_{0,2}$: No difference between dose 2 and the control.

Suppose both these hypotheses are tested at significance level α .

Then, if both null hypotheses are true,

$$Pr\{\text{Reject at least one of } H_{0,1} \text{ and } H_{0,2}\} > \alpha.$$

Outline of talk

1. More examples of multiple hypotheses
2. Familywise type I error rate
3. Bonferroni adjustment
4. A Bonferroni procedure with recycling
5. A gatekeeping procedure
6. Closed testing procedures
7. Example: A special sub-population
8. A fallback procedure
9. A large, general class of procedures
10. Testing multiple hypotheses in adaptive and group sequential trial designs

1. More examples

It is easy to create a situation with very many null hypotheses.

Suppose we combine

2 dose levels and 2 methods of treatment delivery,

Patient subgroups defined by gender,

Patient subgroups defined by presence or absence of a biomarker,

3 different endpoints.

So, we could consider

Benefit of treating at the low dose for men with the biomarker present,

Benefit to women when the drug is delivered by injection, using the third endpoint to assess the patient outcome.

... ..

2. The familywise error rate

Suppose we have k null hypotheses, $H_{0,i}: \theta_i \leq 0$ for $i = 1, \dots, k$.

A testing procedure yields a decision to accept or reject each of the k hypotheses.

The procedure's **familywise error rate** under a set of values $(\theta_1, \dots, \theta_k)$ is

$$Pr\{\text{Reject } H_{0,i} \text{ for some } i \text{ with } \theta_i \leq 0\} = Pr\{\text{Reject any true } H_{0,i}\}.$$

The familywise error rate is controlled strongly at level α if this error rate is at most α for all possible combinations of θ_i values. Then

$$Pr\{\text{Reject any true } H_{0,i}\} \leq \alpha \quad \text{for all } (\theta_1, \dots, \theta_k).$$

Using such a procedure, the probability of choosing to focus on the parameter θ_{i^*} and then falsely claiming significance for null hypothesis H_{0,i^*} is at most α .

3. Bonferroni adjustment

Carlo Bonferroni (1892–1960) followed a career in probability and economics at universities in Turin, Bari, and Florence.

His name is associated with a simple — but very useful — result.

Suppose we have k null hypotheses and test each one at significance level α/k .

Then, if all k null hypotheses are true,

$$Pr\{\text{Reject at least one of } H_{0,1} \dots H_{0,k}\} \leq \alpha.$$

If only some of the k null hypotheses are true,

$$Pr\{\text{Reject at least one true } H_{0,i}\} \leq \alpha.$$

So we have **strong control** of the **familywise error rate**.

Using the Bonferroni adjustment

Example

A trial compares a new treatment against control with respect to two endpoints

Endpoint 1: Core MACE,

Endpoint 2: Expanded MACE.

One-sided type I error probability $\alpha = 0.025$ is divided between the endpoints.

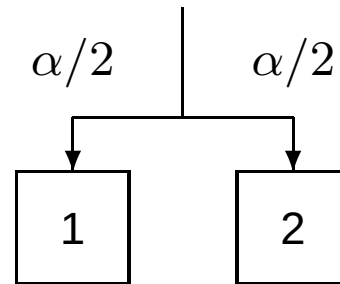
With Z -statistics Z_1 and Z_2 for endpoints 1 and 2,

An effect on Core MACE is declared if $Z_1 > \Phi^{-1}(1 - \alpha/2) = 2.24$,

An effect on Expanded MACE is declared if $Z_2 > \Phi^{-1}(1 - \alpha/2) = 2.24$.

Example continued

This Bonferroni procedure can be represented graphically as:



Familywise type I error is protected conservatively as there is a positive correlation between the two tests, due to the common aspects of the two endpoints.

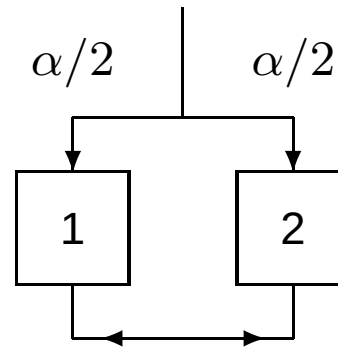
Suppose we have rejected $H_{0,1}$, could it be permissible to test $H_{0,2}$ at significance level α rather than $\alpha/2$?

If $H_{0,1}$ is false, we only need to worry about type I errors concerning $H_{0,2}$.

If $H_{0,1}$ is true, we have already made a type I error, so it will not increase the familywise error rate if we make another!

4. Bonferroni procedure with recycling

We can draw a new version of the Bonferroni procedure, which “recycles” error probability after rejecting $H_{0,1}$ or $H_{0,2}$, as:



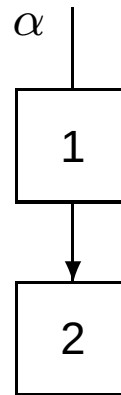
Further options might be to

Divide α unequally between the two hypotheses,

Place the two hypotheses in sequence, so we only consider testing $H_{0,2}$ after rejecting $H_{0,1}$ — this is a limiting case of the above procedure as the division of α tends towards α for $H_{0,1}$ and zero for $H_{0,2}$.

5. A hierarchical testing or “gatekeeping” procedure

We test $H_{0,1}$ first at significance level α . If $H_{0,1}$ is rejected, we continue to test $H_{0,2}$ at significance level α .



To check that familywise type I error rate is protected, consider separate cases

Both $H_{0,1}$ and $H_{0,2}$ are true:

A type I error is committed (and maybe a second) if we reject $H_{0,1}$ — OK.

Only $H_{0,1}$ is true — OK.

Only $H_{0,2}$ is true — OK.

Complex procedures

As we add more options, and get more creative, we can produce some quite complex procedures.

It is still necessary to check that the familywise type I error rate is protected.

At the same time, we should try to avoid excessive conservatism.

We also want to be able to construct testing procedures that fit with:

A logical sequence for considering hypotheses, e.g., primary endpoint before secondary endpoint,

The relative impact of decisions to reject different hypotheses,

The perceived chance of being able to reject each hypothesis.

6. Closed testing procedures

Closed testing procedures (Marcus et al, *Biometrika*, 1976) provide strong control by combining level α tests of each $H_{0,i}$ and of intersections of these hypotheses.

Suppose we have k null hypotheses, $H_{0,i}: \theta_i \leq 0$ for $i = 1, \dots, k$.

For each subset I of $\{1, \dots, k\}$, define the intersection hypothesis

$$H_I = \cap_{i \in I} H_{0,i}$$

which states that all hypotheses $H_{0,i}$ are true, for $i \in I$.

Construct a level α test of each intersection hypothesis H_I , i.e., a test which rejects H_I with probability at most α whenever all hypotheses specified in H_I are true.

The closed testing procedure

The simple hypothesis $H_{0,j}: \theta_j \leq 0$ is rejected overall if, and only if, H_I is rejected for every set I containing index j .

Closed testing procedures

Proof of strong control of familywise error rate

Let $\tilde{I}$ be the set of indices of all true hypotheses $H_{0,i}$.

For a familywise error to be committed, $H_{\tilde{I}}$ must be rejected.

Since $H_{\tilde{I}}$ is true,

$$Pr\{\text{Reject } H_{\tilde{I}}\} = \alpha.$$

Thus,

$$Pr\{\text{Reject } H_{0,j} \text{ for at least one } j \in I\} \leq Pr\{\text{Reject } H_{\tilde{I}}\} = \alpha,$$

so the probability of a familywise error is no greater than α .

7. Example: A special sub-population

Consider a new treatment developed to disrupt a disease's biological pathway.

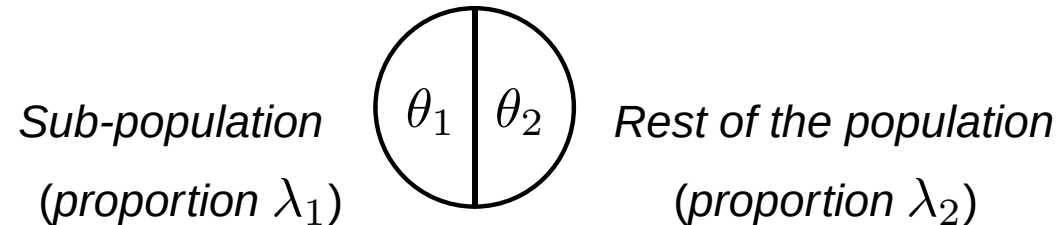
Patients with high levels of a biomarker associated with this pathway should gain particular benefit, but the treatment's wider action may help the general patient population.

As an example, it is recognised that only a portion of the patient population appears to respond to some current cancer treatments.

For new therapies, a target population may be specified — and also a smaller sub-population, in which the treatment is expected to be particularly effective.

Later on, we shall look at “enrichment designs” which adapt a trial design as information accrues about the treatment effects in the full population and sub-population. For now, we consider a non-adaptive trial design.

A special sub-population



Denote the treatment effect:

In the sub-population by θ_1 ,

In the complement of the sub-population by θ_2 ,

Aggregated over the whole population by $\theta_3 = \lambda_1\theta_1 + \lambda_2\theta_2$.

The null hypothesis for the sub-population is $H_{0,1}: \theta_1 \leq 0$.

The null hypothesis for the full target population is $H_{0,3}: \theta_3 \leq 0$.

Sets $H_{0,1}$ and $H_{0,3}$ are not nested (we expect $\theta_1 > \theta_3$ but this may not be so).

All tests are against one-sided alternatives, $\theta_1 > 0$ or $\theta_3 > 0$.

Closed testing procedure for the sub-population example

To apply the closed testing procedure, we need level α tests for the hypotheses

$$H_{0,1}$$

$$H_{0,3}$$

$$H_{0,13} = H_{0,1} \cap H_{0,3}.$$

Suppose we can produce P-values P_1 and P_3 for tests of $H_{0,1}$ and $H_{0,3}$.

So under $H_{0,1}$

$$P\{P_1 \leq x\} \leq x, \quad \text{for } 0 \leq x \leq 1$$

and under $H_{0,3}$

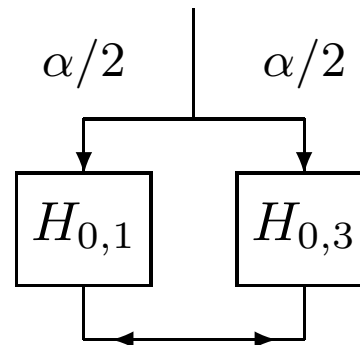
$$P\{P_3 \leq x\} \leq x, \quad \text{for } 0 \leq x \leq 1.$$

We could use a Bonferroni rule and reject $H_{0,13}$ if

$$\min(P_1, P_3) < \alpha/2.$$

Closed testing procedure for the sub-population example

Using the Bonferroni rule, with equal weights, to test $H_{0,13}$ turns out to give the same overall scheme as the “Bonferroni procedure with recycling” seen earlier.

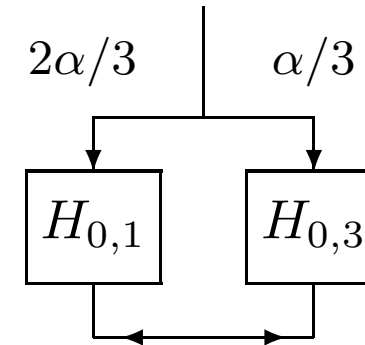


Cases to consider:

	<i>Individual hypothesis tests</i>	<i>Overall decisions</i>
$P_1 < \alpha/2, P_3 < \alpha$	Reject $H_{0,1}, H_{0,3}, H_{0,13}$	Reject $H_{0,1}, H_{0,3}$
$P_1 < \alpha, P_3 < \alpha/2$	Reject $H_{0,1}, H_{0,3}, H_{0,13}$	Reject $H_{0,1}, H_{0,3}$
$P_1 < \alpha/2, P_3 > \alpha$	Reject $H_{0,1}, H_{0,13}$	Reject $H_{0,1}$
$P_1 > \alpha, P_3 < \alpha/2$	Reject $H_{0,3}, H_{0,13}$	Reject $H_{0,3}$

8. Closed testing procedure for the sub-population example

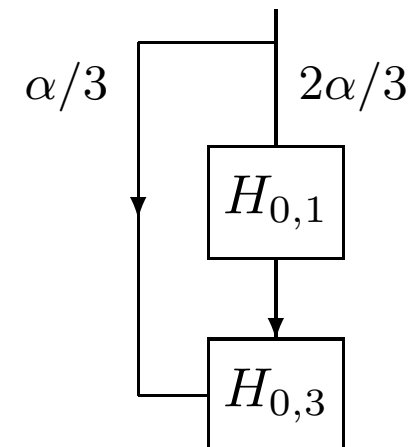
If we suspect there may only be a strong treatment effect in the sub-population, we could use unequal weights in the Bonferroni rule for testing $H_{0,13}$.



This is similar to a “fallback” procedure.

Here, error probability passes from $H_{0,1}$ to $H_{0,3}$ but some is allocated directly to $H_{0,3}$, in case $H_{0,1}$ is not rejected.

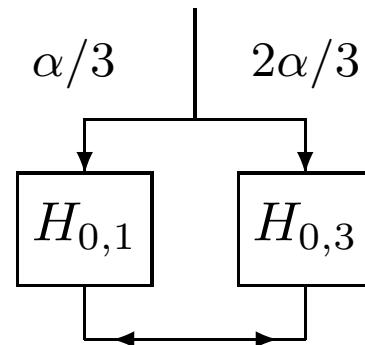
The opportunity to recycle probability from $H_{0,3}$ to $H_{0,1}$ is not used.



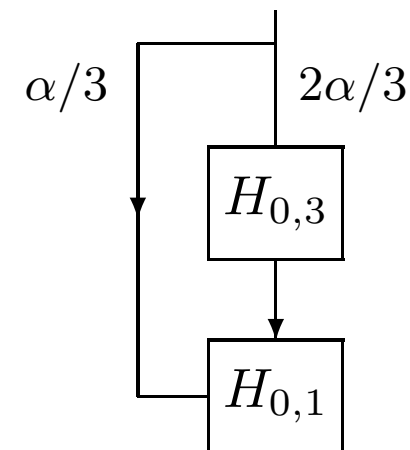
Closed testing procedure for the sub-population example

If we believe there is likely to be a substantial treatment effect in the full population, we could reverse the positions of $H_{0,1}$ (sub-population) and $H_{0,3}$ (full population).

Weighted Bonferroni
with recycling



Fallback procedure



We could go further in this direction and allocate all of α to $H_{0,3}$ initially.

But then we would only start testing $H_{0,1}$ after rejecting $H_{0,3}$ — and establishing there is a treatment effect in the full population.

9. General methodology

Two papers, published simultaneously, describe an elegant and understandable way to describe complex multiple testing procedures.

These are:

“A recycling framework for the construction of Bonferroni-based multiple tests” by Burman, Sonesson and Guilbaud, *Statistics in Medicine*, 2009.

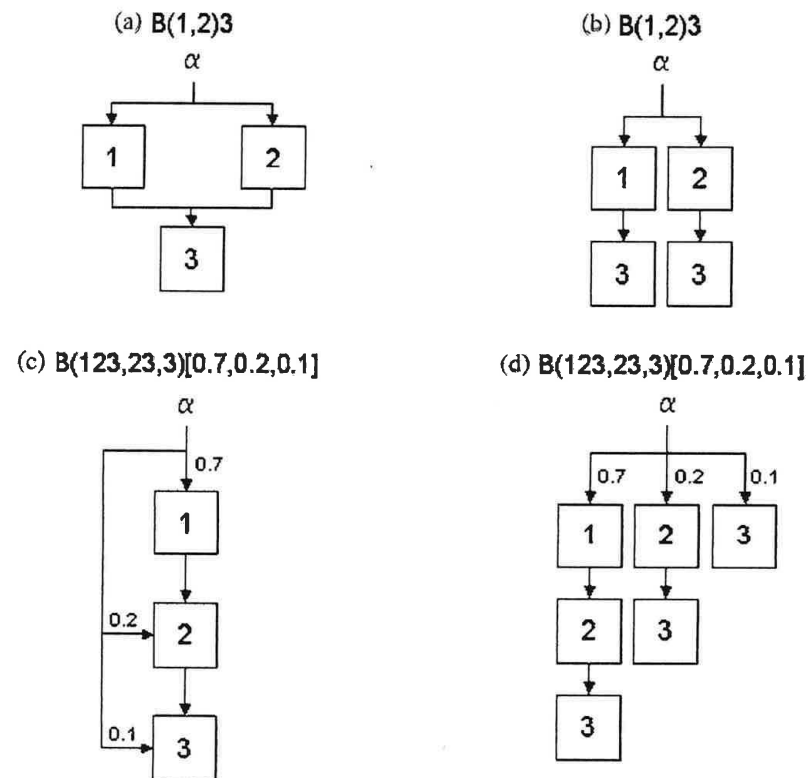
and

“A graphical approach to sequentially rejective multiple test procedures” by Bretz, Maurer, Brannath and Posch, *Statistics in Medicine*, 2009.

A figure from Burman et al. (2009)

(a) and (b) A parallel gatekeeping procedure (equivalent versions)

(c) and (d) A fallback procedure (equivalent versions)



A figure from Bretz et al. (2009)

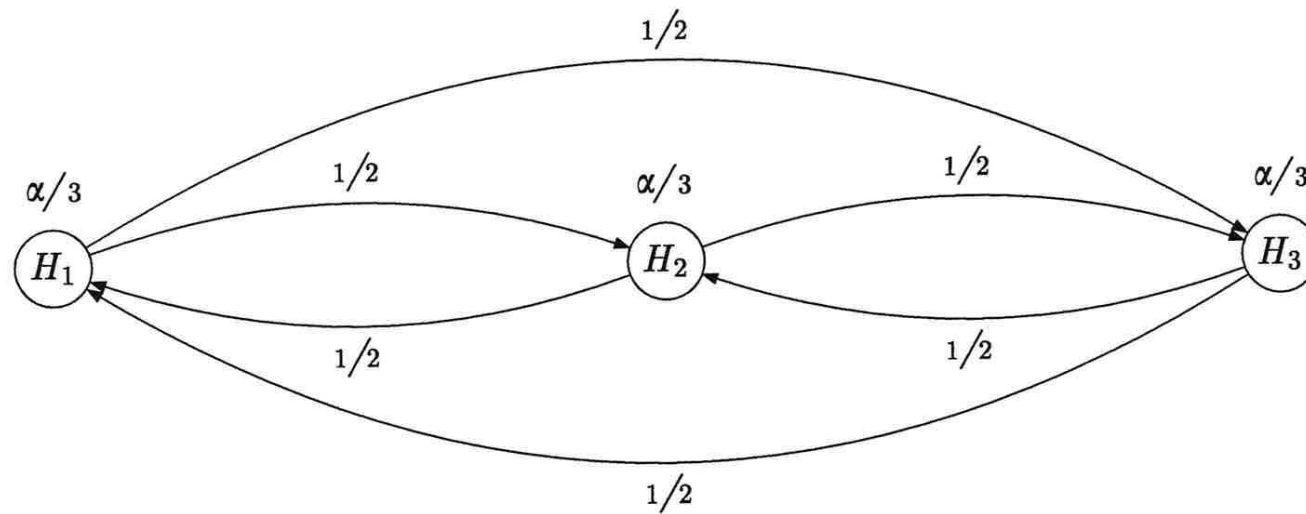


Figure 3. Graphical illustration of the Bonferroni–Holm procedure with $m = 3$ hypotheses and initial allocation $\alpha = (\alpha/3, \alpha/3, \alpha/3)$.

10. Multiple testing procedures and adaptive trial designs

One reason for adaptation is the selection of hypotheses to focus on as a trial progresses.

In the sub-population example, we could use interim data to estimate the treatment effect in the sub-population and full population.

These estimates could inform a decision to continue as planned, or to restrict attention to the sub-population and recruit more patients of this type (enrichment).

If several doses are compared to a control in a Phase III trial, one of the doses may be dropped at an interim analysis.

Thus, multiple testing procedures will be part of the statistical analysis plan.

We shall look later at how these procedures can be combined with the “combination tests” commonly used in adaptive designs.

Multiple testing procedures and group sequential designs

We have described multiple testing procedures in the context of a fixed sample size trial design.

Here, the sample space is simple and it is straightforward to assign a P-value to a given hypothesis.

In principle, this methodology can be applied in a group sequential setting — but we need to have a proper definition for each P-value.

In particular, this definition must not change in response to observed data.

These considerations underlie discussion in the paper

“Statistical considerations for testing multiple endpoints in group sequential or adaptive clinical trials” by Hung, Wang and O’Neill, *J. Biopharmaceutical Statistics*, 2007.

Multiple testing procedures and group sequential designs

Suppose a trial has two endpoints and is run group sequentially.

At the outset, attention focuses on the primary endpoint and the relevant null hypothesis is tested at level α , using a group sequential design.

Following a gatekeeping approach, the secondary endpoint will be tested if the null hypothesis for the primary endpoint is rejected. At this point, a level α “test” will be used for the secondary endpoint.

Suppose the null hypothesis for the primary endpoint is rejected at the final analysis (number 5 out of 5). Can we simply apply a fixed sample test of the secondary endpoint at significance level α ?

Had we rejected the null hypothesis for the primary endpoint at analysis 3 of 5, we would have started testing the secondary endpoint then — but the definition of the test of a hypothesis should not change in response to observed data.

Multiple testing procedures and group sequential designs

What we need in this example, is a group sequential test for the secondary endpoint which gives critical values at each analysis, whether or not we are in a position to use them (depending on the test for the primary endpoint). (*)

Then, crossing this boundary really does imply a P-value less than α for the null hypothesis.

We can explain the high critical values in a group sequential test, by saying we have to control the multiple opportunities to make a type I error.

It would be just the same if we performed one analysis — but chose when to conduct it !

Choosing when to test endpoint B based on observed measurements of endpoint A can bias the test of B if there are correlations between endpoints.

Departing from proven procedures can inflate the type I error rate for (at least) some hypotheses — so follow (*).

Conclusions

1. There are methods available to test multiple hypotheses AND control the overall type I error probability.
2. There are many options to choose from. A suitable choice will depend on the importance to investigators of rejecting each null hypothesis and the likelihood of each null hypothesis being true or false.
3. Closed testing procedures encompass a variety of useful types of multiple hypothesis test.
4. Graphical representations (SiM papers, 2009) can help investigators to select — and understand — an appropriate procedure.
5. Additional care is needed in using these methods in combination with adaptive or group sequential clinical trial designs.